How-To Library

Informatica

Address Verification 6.x Best Practices for Germany

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Abstract

Supported Versions

- Informatica Address Verification 6.4
- Informatica Address Verification (On-Premises) 6.4

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Introduction

Informatica Address Verification can validate and enhance addresses in Germany. This how-to article provides you with configuration information and best practices that you can implement when you verify addresses in Germany. The best practices include advice on how to select the appropriate JSON elements for Germany address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to identify the different types of address information in the input data. You must map each item of address information to an appropriate element in Address Verification.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

Process Modes

Informatica Address Verification supports the following process modes for Germany addresses:

- ADDRESS CODE LOOKUP
- BATCH
- GEOCODE TO ADDRESS
- INTERACTIVE
- QUICK CAPTURE

Character Set

Germany reference address databases store address data in the Latin-1 script. By default, Informatica Address Verification returns the verified Germany addresses in the Latin-1 script.

The Latin-1 script includes ü, ä, ö, and ß characters.

Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The Germany postal service recognizes the following structure for a postal address:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: [STREET] [HOUSE NUMBER]
Line 3: [POSTAL CODE] [CITY]
Line 4: COUNTRY
```

The following sample address illustrates the structure:

```
John Smith
Röntgenstr. 9
67133 MAXDORF
GERMANY
```

Key Address Elements

Address Verification is unlikely to verify an address successfully if one of the following items is absent from the address:

- House Number
- Street
- Locality
- Postal Code

Mapping Input Address Data to Address Elements

To verify an address, map the input address information to appropriate elements in Address Verification.

The following table shows how the input address data maps to the Address Verification address elements:

Information Type	Address Information Example	Address Element
Street, House Number	`Röntgenstr. 9`	Street, HouseNumber
Postal Code	`67133`	PostalCode
City	`MAXDORF`	Locality
Country	`GERMANY`	Country

Input and Result JSON in Address Verification (On-Premises)

The following example shows how you can map the address information to the JSON address elements in Address Verification (On-Premises).

You enter the following input address:

```
{
  "Parameters": {
    "Mode": "Batch",
    "CountryDetermination": {},
    "DataSetDetermination": {}
  },
  "IO": {
    "Inputs": [
      {
        "AddressElements": {
          "Street": [ { "Value": "Jungholz" } ],
          "HouseNumber": [ { "Value": "8" } ],
          "Locality": [ { "Value": "Jungholz" } ],
          "PostalCode": [ { "Value": "87491" } ],
          "Country": "DEU"
        }
      }
    ]
  }
}
```

Informatica Address Verification (On-Premises) returns the following result:

```
{
  "Parameters": {
    "Mode": "Batch",
    "CountryDetermination": {},
    "DataSetDetermination": {}
  },
  "IO": {
    "Inputs": [
      {
        "AddressElements": {
          "Street": [ { "Value": "Jungholz" } ],
          "HouseNumber": [ { "Value": "8" } ],
          "Locality": [ { "Value": "Jungholz" } ],
          "PostalCode": [ { "Value": "87491" } ],
          "Country": "DEU"
        }
      }
    ],
    "Outputs": [
      {
        "ResultInfo": {
          "ProcessStatus": "V",
          "ProcessModeUsed": "Batch",
          "DeliverAdditionalSuggestions": "No",
          "ResultCount": 1,
          "ResultCountOverflow": false,
          "ResultCountries": [ "DEU" ]
        },
        "Results": [
          {
            "Variants": [
              {
                "StatusValues": {
                  "AddressType": "S",
                  "LanguageISO3": "DEU",
                  "UsedVerificationLevel": "All",
                  "MatchPercentage": "100.00",
                  "Script": "Native Primary",
                  "AddressCount": "",
                  "ResultQuality": 6
                },
```

```
                "AddressElements": {
                  "Street": [ { "Value": "Jungholz" } ],
                  "HouseNumber": [ { "Value": "8" } ],
                  "Locality": [ { "Value": "Jungholz" } ],
                  "PostalCode": [ { "Value": "87491" } ],
                  "Country": [ { "Code": "DEU", "Name": "GERMANY" } ]
                },
                "PreformattedData": {
                  "PostalDeliveryAddressLines": [ { "Value": "Jungholz 8" } ],
                  "PostalFormattedAddressLines": [
                    { "Value": "Jungholz 8" }, { "Value": "87491 Jungholz" }
                  ],
                  "PostalLocalityLine": { "Value": "87491 Jungholz" }
                }
              }
            ]
          }
        ]
      }
    ]
  }
}
```

Input and Result JSON in Address Verification

The following example shows how you can verify a Germany address in a REST call to Address Verification.

You submit the following request:

```
{
  "Login": *****,
  "Password": *****,
  "CampaignId": "Campaign222",
  "UseTransactions": "TEST",
  "Request": {
    "Parameters": {
      "Mode": "Batch",
      "CountryDetermination": {},
      "DataSetDetermination": {},
      "CountrySets": [
        {
          "Process": {
            "AlternativeHandling": {
              "OutdatedAddresses": { "Match": "Off", "GetNewAddress": false },
              "AliasAddresses": { "Match": "Off" }
            },
            "VerificationLevel": {
              "PreferredVerificationLevel": "All",
              "MinimumVerificationLevel": "All"
            },
            "Geocode": { "SearchRadius": "50" },
            "Enrichments": {
              "Geocoding": {
                "Rooftop": false,
                "ArrivalPoint": false,
                "StreetCenter": false,
                "PostalCodeCenter": false,
                "LocalityCenter": false
              },
              "Global": false,
              "CountrySpecific": false,
              "CAMEO": false
            }
          }
        }
      ]
    },
  "IO": {
    "Inputs": [
      {
```

```
          "AddressElements": {
            "Street": [ { "Value": "Jungholz" } ],
            "HouseNumber": [ { "Value": "8" } ],
            "Locality": [ { "Value": "Jungholz" } ],
            "PostalCode": [ { "Value": "87491" } ],
            "Country": "DEU"
          }
        }
      ]
    }
    }
}
```

Informatica Address Verification returns the following response:

```
{
    "Status": "Ok",
    "StatusDescription": "No error",
    "TransactionInfo": {
        "TotalTransactionsCharged": 1,
        "TransactionPools": [
            {
                "Type": "BATCH",
                "TransactionsCharged": 1
            }
        ]
    },
    "CampaignId": "Campaign222",
    "Response": [
        {
            "ResultInfo": {
                "ProcessStatus": "V",
                "ProcessModeUsed": "Batch",
                "DeliverAdditionalSuggestions": "No",
                "ResultCount": 1,
                "ResultCountOverflow": false,
                "ResultCountries": [
                    "DEU"
                ]
            },
            "Results": [
                {
                    "Variants": [
                        {
                            "StatusValues": {
                                "AddressType": "S",
                                "LanguageISO3": "DEU",
                                "UsedVerificationLevel": "All",
                                "MatchPercentage": "100.00",
                                "Script": "Native Primary",
                                "AddressCount": "",
                                "ResultQuality": 6
                            },
                            "AddressElements": {
                                "Street": [
                                    {
                                        "Value": "Jungholz"
                                    }
                                ],
                                "HouseNumber": [
                                    {
                                        "Value": "8"
                                    }
                                ],
                                "Locality": [
                                    {
                                        "Value": "Jungholz"
                                    }
                                ],
                                "PostalCode": [
                                    {
```

```
                        "Value": "87491"
                    }
                ],
                "Country": [
                    {
                        "Code": "DEU",
                        "Name": "GERMANY"
                    }
                ]
            },
            "PreformattedData": {
                "PostalDeliveryAddressLines": [
                    {
                        "Value": "Jungholz 8"
                    }
                ],
                "PostalFormattedAddressLines": [
                    {
                        "Value": "Jungholz 8"
                    },
                    {
                        "Value": "87491 Jungholz"
                    }
                ],
                "PostalLocalityLine": {
                    "Value": "87491 Jungholz"
                }
            }
        }
    ]
  }
]
}
```


Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified Germany addresses.

The following table lists the types of enrichment that you can add to a Germany address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments for valid Germany addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Standard Geocoding	You can retrieve the standard, interpolated geocoordinates for valid Germany addresses.
Arrival Point Geocoding	You can retrieve the arrival point geocoordinates for valid Germany addresses.
Rooftop Geocoding	You can retrieve the rooftop geocoordinates for Germany addresses that have the rooftop geocoordinates.

Address Verification Tips and Best Practices

Note the following points when you verify Germany addresses:

- Address Verification does not support address information in `FormattedAddressLine` or `DeliveryAddressLine` elements in QuickCapture mode for Germany addresses.

- The building and sub-building information in a Germany address follows the house number information. An address in Germany uses a forward-slash symbol to separate building or sub-building information from house number information. For example, `Bahnhofstr. 12/3`.
- Address Verification verifies alphanumeric house number information such as `5 A` and `8 C` in a Germany address.

  Note: Each alphanumeric house number is based on a numeric house number. For example, a street with a house number 6 A also has a house number 6.

  Address Verification returns a V4 status for both of the following addresses in batch and interactive modes:

```
Fahrenkuhl 6
Kronsburg
Kiel 24145
Schleswig-Holstein
```

```
Fahrenkuhl 6A
Kronsburg
Kiel 24145
Schleswig-Holstein
```

Reference Address Databases

Informatica updates some of the Germany reference address databases every month and some every quarter. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the Informatica DaaS Newsletter from Informatica.

Informatica Address Verification provides the following reference databases for Germany every month:

- `DEU_ADV_VRF_BIA_000_6_x_0.MD6`. Batch and interactive data.
- `DEU_ADV_RLK_ACL_000_6_x_0.MD6`. Address code lookup data.
- `DEU_ADV_VRF_QCP_000_6_x_0.MD6`. QuickCapture data.
- `DEU_ADV_ENR_EN1_000_6_x_0.MD6`. Enrichment data.

Informatica Address Verification provides the following reference databases for Germany every quarter:

- `DEU_ADV_RLK_GTA_000_6_x_0.MD6`. GeocodeToAddress data .
- `DEU_ADV_ENR_GST_000_6_x_0.MD6`. Standard geocoding data.
- `DEU_ADV_ENR_GAP_000_6_x_0.MD6`. Arrival point geocoding data.
- `DEU_ADV_ENR_GRT_000_6_x_0.MD6`. Rooftop geocoding data.

Note: You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification runs on Informatica Cloud.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://docs.informatica.com/data-as-a-service/address-verification-(on-premises)/6-4-0.html

https://docs.informatica.com/data-as-a-service/address-verification/6-4-0.html

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